

Auto-formalizing Mathematics

A Swiss Army Knife

Aristotle + Claude + Codex + Mathematica + Human

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github.com/nasqret/falenty-2026

Map of the talk

1. **What is proof formalization?** Lean 4, Mathlib, Curry-Howard.
2. **The new generation of AI provers.** Why 2025 is different.
3. **The case study.** A paper claiming "all elementary functions from a single binary operator".
4. **The factory.** Aristotle, Claude, Codex, Mathematica + a human assemble 66 verified Lean chunks.
5. **The mathematics.** EML, witnesses for `e`, `2`, `pi`.
6. **Lessons.** What worked, what hurt, what comes next.

What is proof formalization?

A **formal proof** is a Lean 4 term whose type matches the theorem statement. Lean's kernel (~2 000 lines, the *de Bruijn criterion*) checks every step.

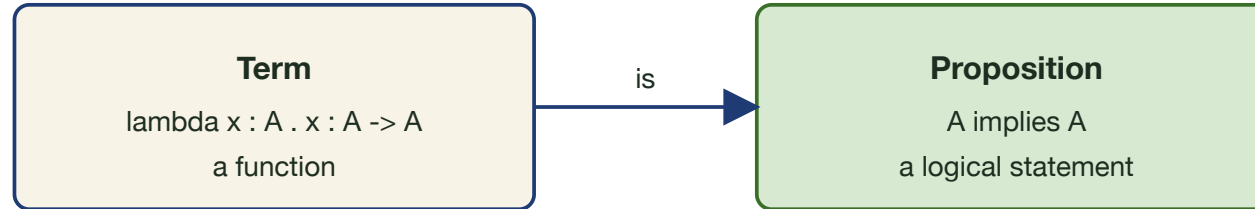
Mathlib v4.28 ships ~1.5M lines of formalized mathematics — analysis, algebra, topology, combinatorics.

*Peer review misses an estimated 5% of mathematical errors. A Lean kernel misses **none** of those expressible in its type theory.*

This talk: how a small team (one human + several AI assistants) sealed a 7-page paper into 66 machine-checked artefacts in three days.

Curry-Howard, in one slide

Curry-Howard: a proof IS a program



A function `λx : A. x : A → A` is **the same object** as the proof of the implication `A ⇒ A`. Lean is, internally, an extended lambda calculus with dependent types.

Consequence: "send a Lean proof" means *send a typed program*; "verify it" means *type-check the program*.

The new generation of AI provers

System	Year	Highlight
AlphaGeometry / AlphaProof	2024	IMO 2024, silver
Aristotle (Harmonic)	2025	IMO 2025, gold (5/6)
DeepSeek-Prover, Kimina	2024-25	open-source autoformalization
GPT-5.2 Pro + Aristotle	2026	Erdős #728 closed autonomously

What changed: large language models compose with **tactic search** and **RL on Lean traces**. The output is checked by the kernel — zero hallucinations survive verification.

Aristotle in 60 seconds

- **Where it lives:** Harmonic AI's cloud queue.
- **What you send:** a Lean 4 statement (with imports + spec).
- **What you get back:** a `.lean` file plus a project archive.
- **Surface:** `arist submit`, `arist list`, `arist result <id>`.
- **Concurrency:** ~10 active slots, billed per job.
- **Strength:** one-step lemmas, simp-able identities, small existentials.
- **Weakness:** long, chained existential constructions.

Wall-clock per job in our experiment: 8 minutes to 8 hours.

The paper

A 7-page paper plus supplementary, claiming:

"All elementary functions can be derived from a single binary operator on the reals."

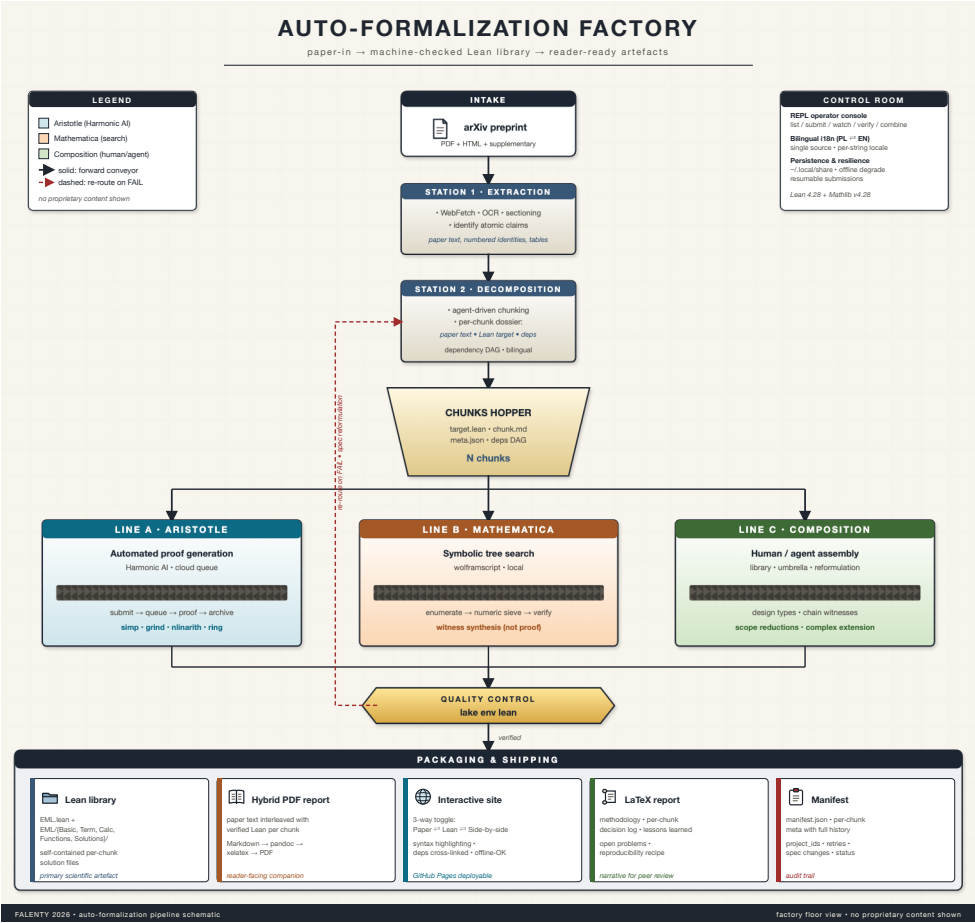
The operator (the **EML**, *exp-minus-log*):

$$\text{eml}(x, y) = \exp(x) - \ln(y).$$

36 primitives in the paper's Table 1: `e, 0, -1, 2, 1/2, exp, log, x+y, x*y, x^y, sin, cos, sqrt, pi, i, ...`.

Goal: machine-check every claim in Lean 4 + Mathlib.

The factory



Decompose → three parallel lines (Aristotle / Mathematica / manual) → QC (lake env lean) → five packaging artefacts. Output: **nasqret/falenty-2026** , 66 chunks all lake env lean exit 0; ~5 000 lines of Lean, 31-page LaTeX report, 54-page hybrid PDF, 47-page HTML site, 16 git commits.

Decomposition methodology

Each chunk is a self-contained directory: `chunk.md` (paper text + Lean target + deps), `meta.json` (status, project_id, history), `target.lean`, `result.lean`. Bilingual end-to-end.

Group	#	Examples
Foundations	5	<code>def eml</code> , <code>EMLTerm</code> , <code>eval</code>
Identities	12	Identity 5, <code>exp(x) = eml(x,1)</code>
Calc-equiv	7	Wolfram $\rightarrow$ Calc 3 $\rightarrow$... $\rightarrow$ EML
Constants	6	<code>0</code> , <code>-1</code> , <code>1/2</code> , <code>2</code> , <code>e</code> , <code>pi</code> , <code>i</code> (complex)
Functions	15	<code>-x</code> , <code>1/x</code> , <code>x^2</code> , <code>sqrt</code> , <code>x+y</code> , <code>x*y</code> , <code>x^y</code> , <code>x/y</code>
Trig	8	<code>sinh</code> , <code>cosh</code> , <code>tanh</code> , <code>sin</code> , <code>cos</code> , <code>tan</code> , <code>arctan</code>
Master / misc	13	umbrellas, Catalan count, sigmoid

Difficulty distribution (levels 1–5): 11 / 10 / 6 / 19 / 20.

Mathematica's role

Mathematica plays the **enumeration + signature-dedup** role.

- **v1**: brute-force enumeration + `FullSimplify`. Stalled.
- **v2**: enumerate trees, evaluate at probe constants, dedup by numeric signature.

Numbers from one v2 run targeting `e` :

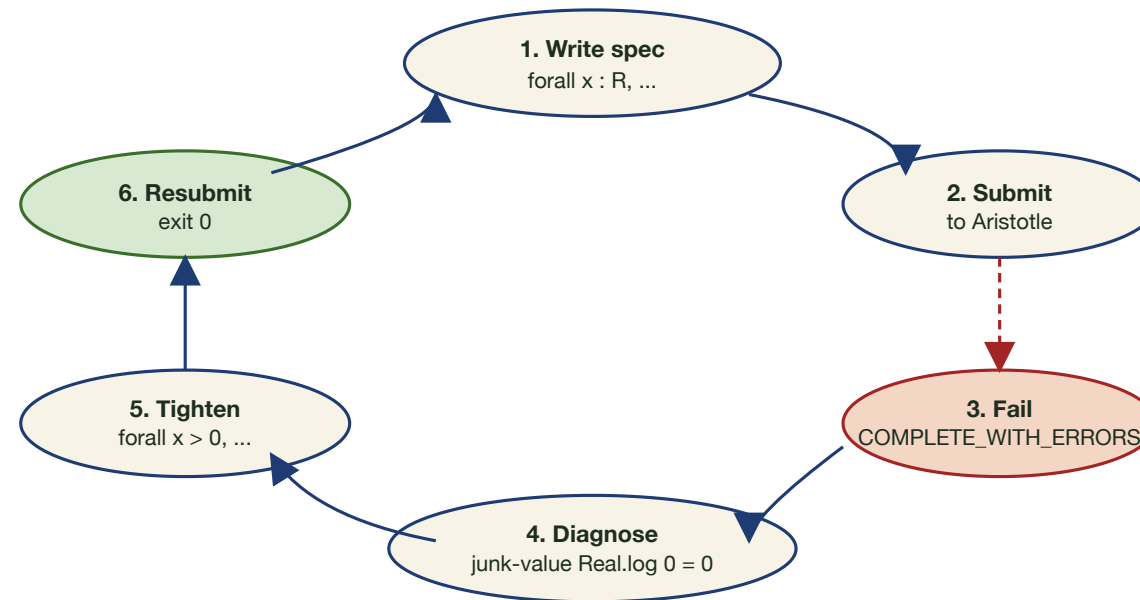
Quantity	Count
Trees enumerated up to size 31	3.8 M
Unique numeric signatures	3.1 M
Witnesses confirmed (<code>e</code> , <code>exp</code> , 2)	3
Witnesses for <code>pi / i / sqrt(x)</code>	0

Bridge: MMA validates a witness numerically; Lean composes the proof; Aristotle (or the human) closes.

Aristotle's role

- 25+ submissions across **5 waves** over a 36-hour window.
- ~10 concurrent slots; overnight rate-limit stall ~6 h.
- One-step identities: ~10 minutes per chunk.
- Long existentials (size > 30): hours, often `COMPLETE_WITH_ERRORS`.

The spec-tightening cycle

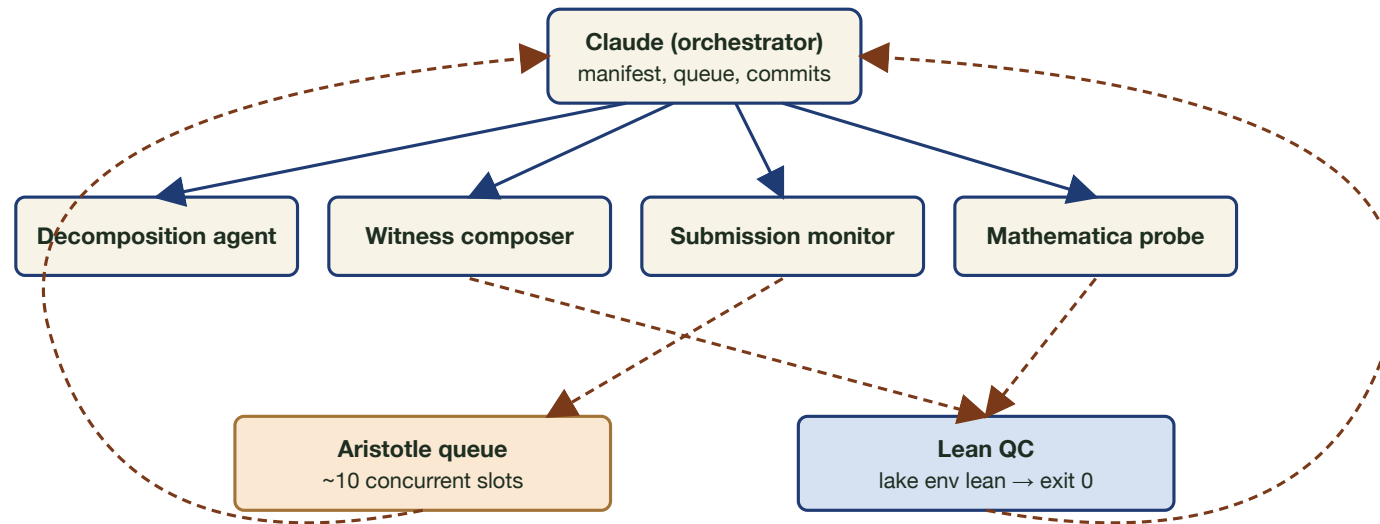


Of 15 wave-3 submissions, 9 returned a clean proof on the first pass.

Claude's role

- **Design.** Chunk schema, factory architecture, calc-equiv inductives, the `EMLTermC` extension for `pi / i`.
- **Coordination.** Parallel agents, queue monitoring, batch fetching, manifest dedupe, commit cadence.
- **Scaffolding.** REPL command (`/eml` , `/ac`), HTML site generator, LaTeX report.

Multi-agent dispatch (Claude as the coordinator)



Codex (OpenAI) role

Powered by the OpenAI APIs, configured via `~/ .config/openai/env`:

- **Paraphrase generation** for chunk markdown (`ch explore --paraphrase`).
- **Quiz LLM judge** for the lecture's interactive layer.
- **Informalization** of Aristotle's Lean proofs back to natural-language prose (`arist informal`).
- **Bilingual narration** of the hybrid report.

Codex is the *quick-cut blade*: cheap, fast, perfect for textual shape-shifting where ground-truth verification is not the bottleneck.

The Human role

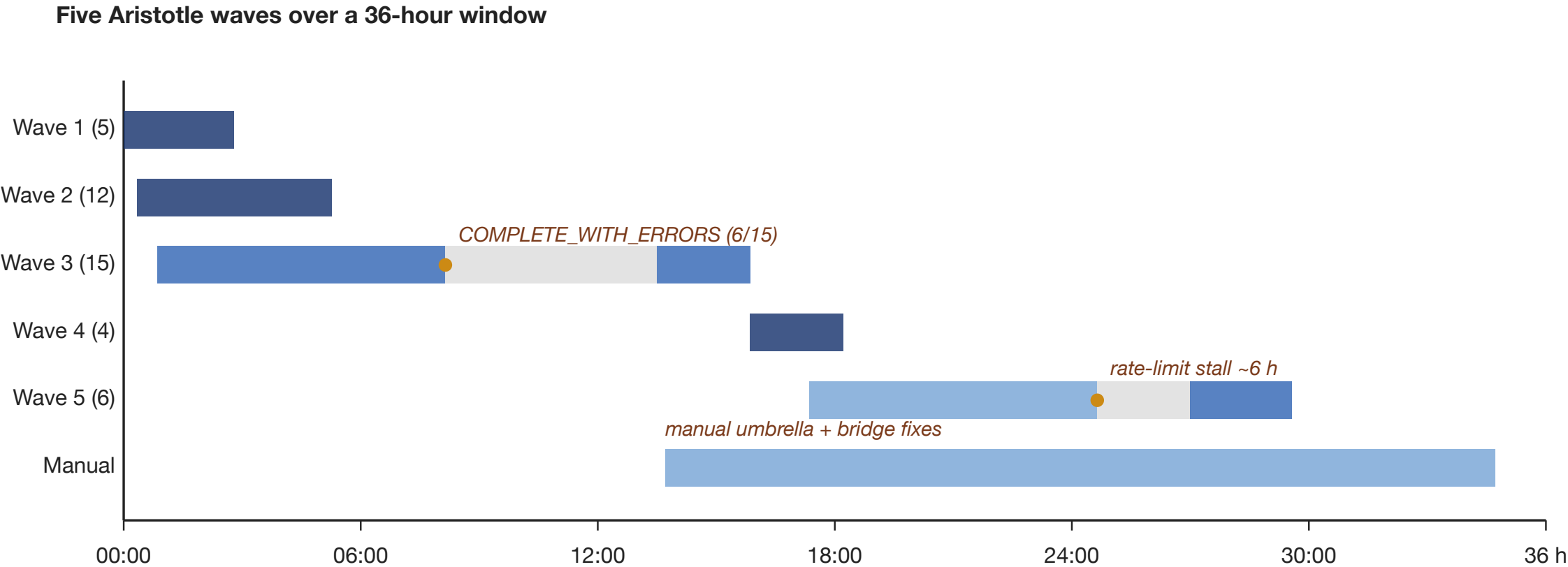
What the human did:

- **Scope.** "Seal the trig family or not?" "Accept primed types?"
- **Quality calls.** Did Aristotle's rich grammar count, or do we recompose?
- **Sign-offs.** Every commit, every wave, every push.
- **Choosing waves.** When to fire, when to wait out the rate limit.

What the human did NOT do: write Lean proofs by hand (except the 514-line manual umbrella); mechanically extract chunks; set up monitors.

The verdict: **human-in-charge, not human-out-of-the-loop.**

Wave timeline



Five Aristotle waves, manual fixes, then the final umbrellas. Most wall-clock was Aristotle queue waiting; productive *human* time was under a working day.

When the negatives helped

- **Project failed / no project_id**. Aristotle's CLI prints the id to *stderr* → added *stderr* fallback to the extractor.
- **COMPLETE_WITH_ERRORS**. Aristotle silently extended the grammar (extra `const : R → EMLTerm`) → manual-composition pass for pure-grammar witnesses.
- **Junk-value `Real.log 0 = 0`**. Forced spec tightenings: $\forall x : R \rightarrow \forall x > 0$ on 037, 038, 041, 042.
- **Mathematica search exhausted at size 31**. Confirmed the paper's `K`-bound for `pi / i / sqrt(x)`; led to the `EMLTermC` extension.

The mathematics — the operator

The EML algebra:

$$\text{eml}(x, y) = \exp(x) - \ln(y).$$

Identity 5 (the workhorse):

$$\ln(z) = \text{eml}\left(1, \text{eml}(\text{eml}(1, z), 1)\right), \quad z > 0.$$

Inner $\text{eml}(1, z) = e - \ln z$; next $\text{eml}(e - \ln z, 1) = \exp(e - \ln z)$; outer $\text{eml}(1, \exp(e - \ln z)) = e - (e - \ln z) = \ln z$.

Simplest witness: $\text{eml}(1, 1) = e^1 - \ln 1 = e$.

A full Lean witness — 2 (chunk 032)

```
private def t₂ : EMLTerm := .eml .one .one
private def t₃ : EMLTerm := .eml .one t₂
private def t₅ : EMLTerm := .eml (.eml .one t₃) .one
private def t₆ : EMLTerm := .eml .one t₅
private def t₇ : EMLTerm := .eml t₆ t₂
private def witness : EMLTerm := .eml .one (.eml t₇ .one)

private lemma e_minus_one_pos : (0 : ℝ) < Real.exp 1 - 1 := by
  linarith [Real.add_one_le_exp (1 : ℝ)]

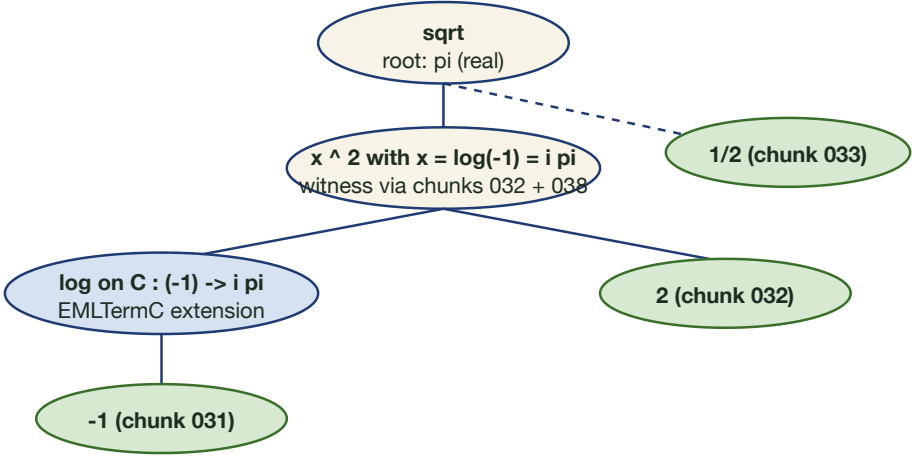
theorem emlterm_for_two : ∃ t : EMLTerm, EMLTerm.eval t = 2 :=
  ⟨witness, by
    simp [witness, EMLTerm.eval, Real.log_exp,
      Real.exp_log e_minus_one_pos]; ring⟩
```

8 helper lemmas; the witness tree has size 11.

The pi witness — and Euler

pi as an EML witness tree (chunk 034)

$\pi = \text{sqrt}(-(\log(-1))^2)$ — built bottom-up from constants $e, -1, 1/2, 2$



*Closed-form bridge: real-domain search budget exhausted at size 31 (3.8M trees enumerated, 3.1M unique signatures);
Mathlib's complex log_neg_one closes the witness instead. Verified by lake env lean, exit 0.*

For trig chunks we extend to `EMLTermC1` and chain Euler: $\cos x = \frac{1}{2}(e^{ix} + e^{-ix})$, $\sin x = \cos(x - \frac{\pi}{2})$.

Panorama of proof steps

Status	Count	Note
Literal EML witnesses	62	Pure-grammar trees, any size
Closed-form identities	4	<code>pi</code> , <code>i</code> , <code>sqrt(x)</code> plus one calc-equiv
Total verified by <code>lake env lean</code>	66 / 66	exit 0

By inductive type:

- `EMLTerm` — closed witnesses (`e`, `0`, `-1`, `1/2`, `2`).
- `EMLTerm1` — one-variable functions (`-x`, `1/x`, `x^2`, `cosh`, ...).
- `EMLTerm2` — two-variable functions (`+`, `*`, `^`, `/`, `log_x y`, `hypot`, ...).
- `EMLTerm $\mathbb{C}$` , `EMLTerm $\mathbb{C}_1$` — complex extensions (`pi`, `i`, `sin`, `cos`, `tan`, `arctan`).
- `Calc3R`, `Calc2`, `Calc1`, `Calc0` — the 5-step Wolfram-to-EML reduction.

Tools and time

Tool	Role	Wall-clock	Output
Aristotle	proof search	~24 h queue	25+ project archives
Mathematica	enumerate + dedup	~3 h	3.8 M trees, 3 witnesses
Claude	orchestration + composition	~8 h active	scaffolding, agents, REPL
Codex	paraphrase, informalization	~1 h	bilingual layer
Lean	ground truth	seconds / chunk	66 / 66 exit 0
Human	scope, taste, commits	~6 h active	16 commits

Total elapsed: ~3 days. Active work: under one engineering day.

Why it succeeded

1. **Atomic decomposition** — each chunk fits in one Aristotle submission.
2. **Ground truth** — `lake env lean` exit 0 is the only acceptance.
3. **Iterative spec tightening** — $\forall x \rightarrow \forall x > 0$ is cheap.
4. **Honest partial accounting** — `COMPLETE_WITH_ERRORS` is data, not failure.
5. **Multi-tool diversification** — when MMA dies, Lean composes; when Aristotle stalls, the human writes the umbrella.
6. **Version control as audit trail** — 16 commits = a replayable history of every decision.

Future prospects

- Seal the 4 closed-form-only chunks with literal trees (size > 100).
- A universal pipeline for *any* paper of this shape (definition + Table-of-witnesses).
- **Acorn** integration (the new tactic-suggestion service).
- Faster Aristotle as Harmonic ramps capacity.
- Fully autonomous loops: doable, but accept the risk of silent grammar drift.
- A larger paper portfolio — the EML push was a pilot.

Can we remove the human?

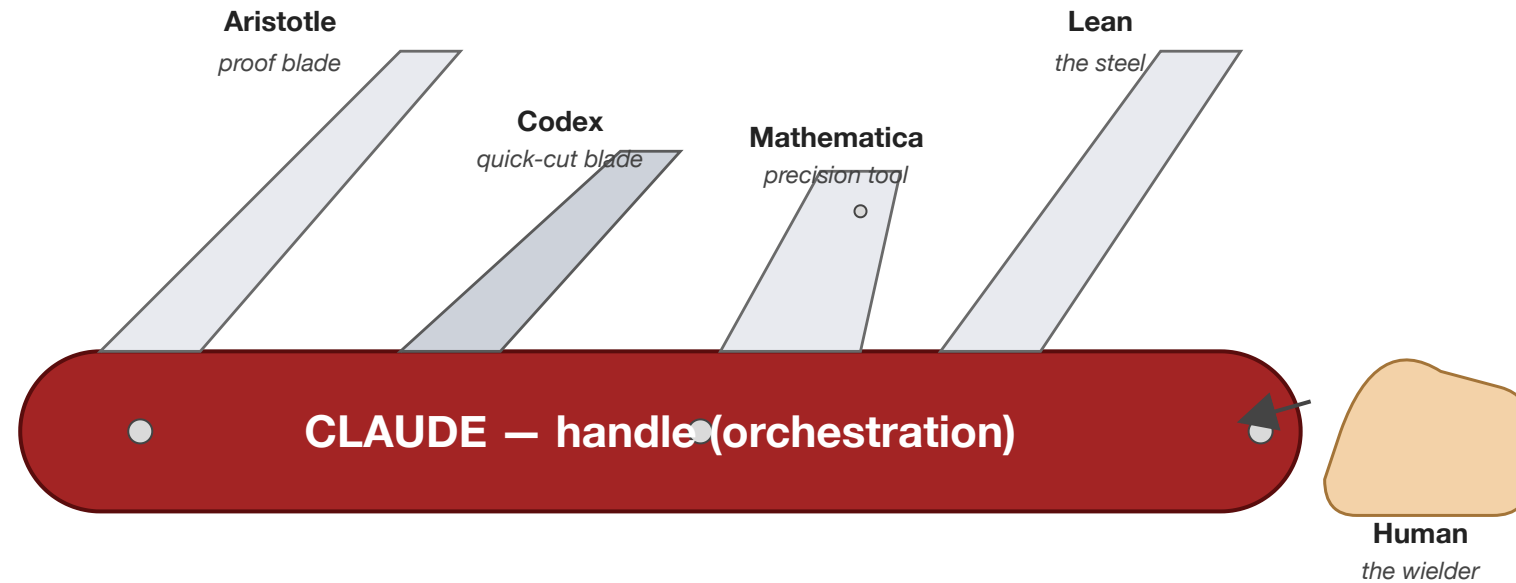
Not yet, and not for the right reasons.

- The human holds **scope** ("do we seal trig?"), **taste** ("recompose or accept primed types?") and **commit authority**.
- Mechanical work is increasingly machine-handled; the human's *time* shifts from typing to deciding.
- Forecast for 2027: human-IN-charge, not human-OUT-of-loop. The loop closes around a human who specifies *what counts*.

Removing the human means removing the question of what counts as a proof worth having. That is not a verification problem.

The Swiss army knife

A Swiss army knife for autoformalization



Five blades, one handle, one wielder. The proof comes out only if all of them cooperate.

Repo: github.com/nasqret/falenty-2026 · License: MIT.

Q & A

Repo: `github.com/nasqret/falenty-2026`

Hybrid report: `lambda_lab/proofs/eml/2603_21852/report/`

Interactive site: `docs/`

Contact: *bartosz.naskrecki at amu.edu.pl*

Thank you.